CHAPTER D1

Solution design and architecture

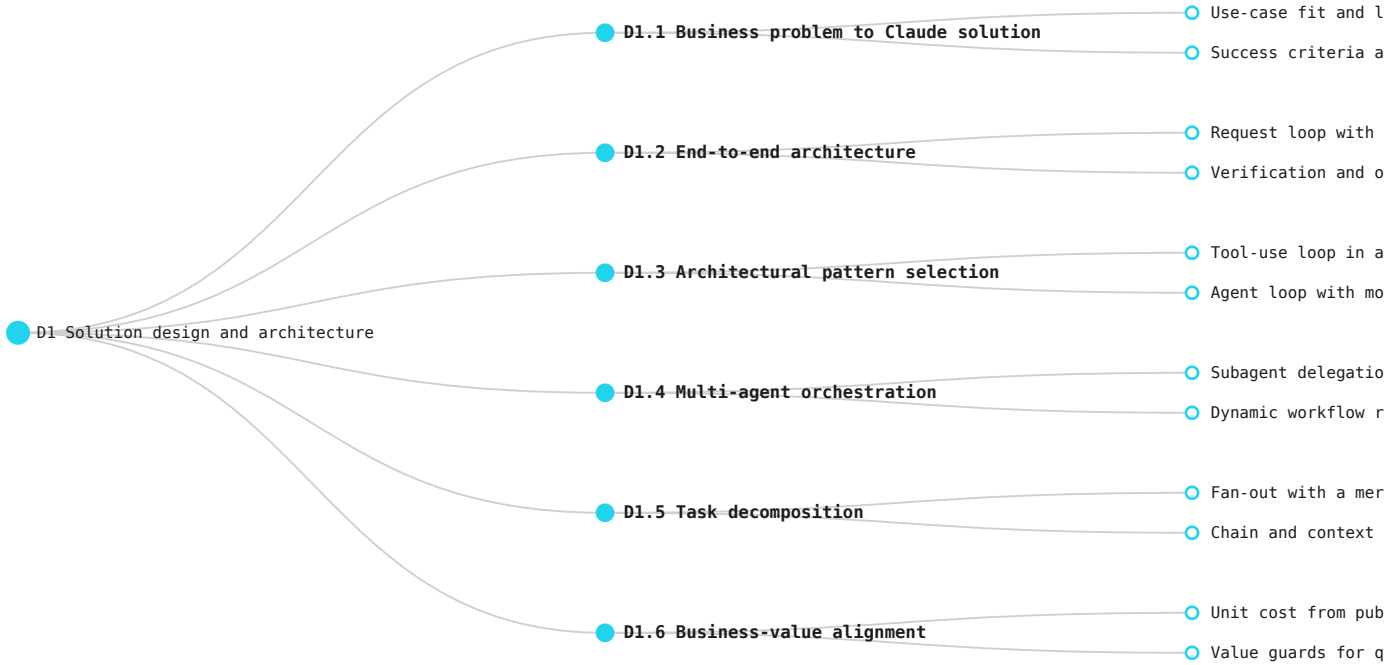
WEIGHT 17%. INFERENCE FROM SECONDARY SOURCES.

Choose the simplest architecture that fits the steps.

6 sections, 12 components, 18 recall cues.

CONCEPT MAP

Units branch to components. On a small screen, scroll sideways.



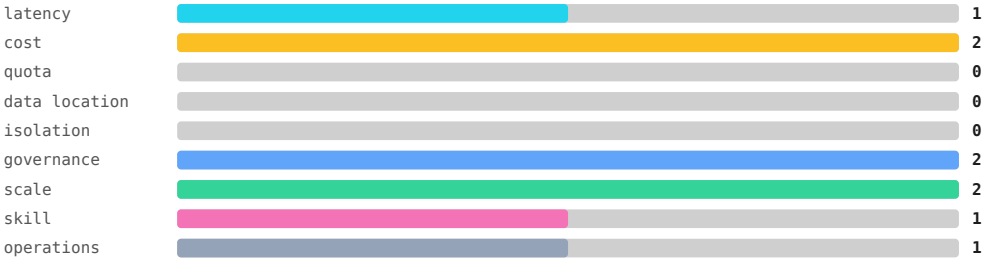
SECTIONS

D1.1 CHAPTER D1, SOLUTION DESIGN AND ARCHITECTURE

Business problem to Claude solution

Fixed mapping: code. Open input: Claude plus a code check.

CONSTRAINT FAMILIES



USE-CASE FIT AND LANE CHOICE

Send fixed mapping to code and send open input to Claude only when a check in code can verify the output. (Inference)

Rule Use code for fixed mapping, and use Claude with a code check for open input.

Trap A model call for a fixed format that a rule covers. It adds cost and a failure mode, and it gives no gain. (Inference)

Exam tip If the stem gives no success metric, add the metric before you pick the architecture. (Inference)

SUCCESS CRITERIA AND EVALS

Write each success criterion as a measurable target with a threshold, and test it on an eval set before the build scales.

Rule Write each criterion as a metric, a threshold, and an owner before you compare models.

Trap A general quality goal, such as good performance. It cannot pass or fail, so no owner can sign it.

Exam tip If an option uses an adjective as the success test, prefer the option that adds a number and a test set. (Inference)

▼ RECALL CUES (3)

1. **reduce cost for a task with a fixed input format**
   Answer code rules over a model call
   Rule Use code for fixed mapping. Call Claude only for open input.
   - Largest model for every input
   - Model call for a field that a rule reads
2. **success is described only with adjectives**
   Answer measurable criteria with threshold and owner
   Rule Add a metric, a threshold, and a named owner before the build.
   - More model tiers with no eval set
   - One demo result used as proof
3. **a wrong output reaches a customer**
   Answer code check on every output
   Rule Place a deterministic check between the model and the user.
   - Prompt wording as the only control
   - Human review of every output with no check

▼ NUMBERS AND VERSIONS (9)

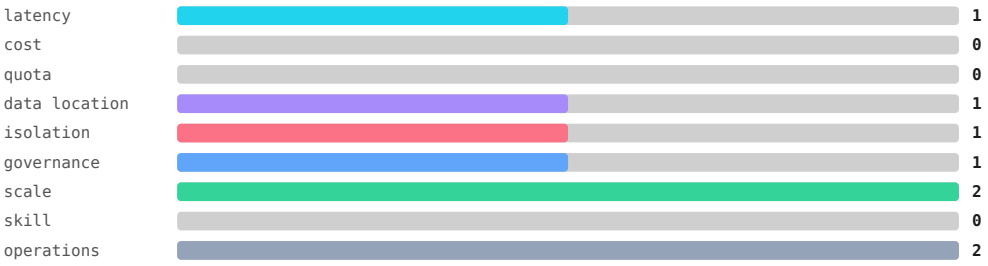
- Claude Haiku 5.5 base input price is 0.10 dollars per million tokens for prompts up to 100,000 tokens. source
- Claude Haiku 5.5 output price is 0.50 dollars per million tokens for prompts up to 100,000 tokens. source
- Claude Haiku 5.5 prompts over 100,000 tokens pay higher prices. source
- Claude Opus 5.5 base input price is 4 dollars per million tokens. source
- The Message Batches API discount is 50 percent on input and output tokens. source
- A batch holds up to 100,000 Message requests or 256 MB, whichever is reached first. source
- The docs give an example criterion of an F1 score of at least 0.85 on a held-out test set. source
- Most batches finish in less than 1 hour. source
- Batch results are available for 29 days after creation. source

D1.2 CHAPTER D1, SOLUTION DESIGN AND ARCHITECTURE

End-to-end architecture

The application owns history, retries, and stop reasons.

CONSTRAINT FAMILIES



REQUEST LOOP WITH STATE IN THE APPLICATION

No state persists between Messages API calls. The application resends the conversation on each call and checks each stop reason.

Rule Keep the history, the retries, and the stop reason handling in application code.

Trap A larger model or a longer prompt for a failure that no trace locates. The change treats a symptom and hides the stage that breaks. (Inference)

Exam tip If a stem shows a failure with no known stage, pick the option that adds a trace per stage. (Inference)

VERIFICATION AND OUTPUT GATES

A check in code compares the draft with the source and the schema before the answer leaves the system. (Inference)

Rule Use a check in code for each output whose error costs money, safety, or trust.

Trap A prompt that asks the model to check its own draft. The check shares the failure mode of the draft, so it gives no independent control. (Inference)

Exam tip If a stem names a schema, pick structured outputs, and add a code check for the values. (Inference)

▼ RECALL CUES (3)

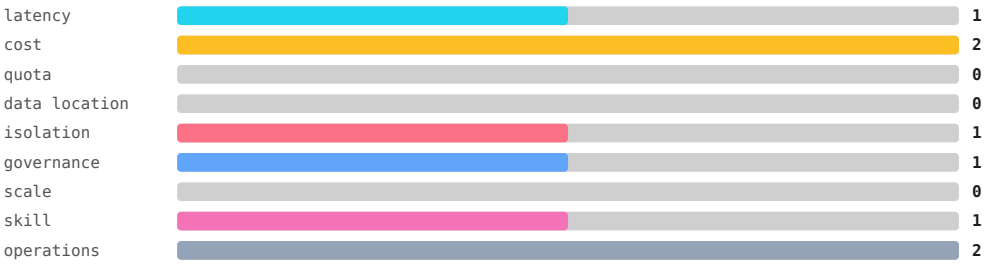
1. **a failure with no known stage**
   Answer per-stage trace before a model change
   Rule Instrument each stage and find the stage that breaks before you change the model.
   - Larger model tier
   - More human reviewers on every output
2. **output must be right for a downstream system**
   Answer schema plus code check
   Rule Use structured outputs and a code check on the values.
   - Prompt that asks for valid JSON only
   - Model that checks its own draft
3. **the conversation must continue across turns**
   Answer application state with full history
   Rule The application holds the history and sends it with each call.
   - Assumed server-side memory between calls
   - History kept only in one browser tab

D1.3 CHAPTER D1, SOLUTION DESIGN AND ARCHITECTURE

Architectural pattern selection

Known steps: tool-use loop. Open steps: agent loop.

CONSTRAINT FAMILIES



TOOL-USE LOOP IN APPLICATION CODE

The model asks for a tool call, and the application runs the tool and returns the result.

Rule Use a tool-use loop when the steps are known, and the model only chooses tools.

Trap A full agent for one judgment step. The agent adds planning overhead that the task does not need. (Inference)

Exam tip If the stem names one tool call and a fixed sequence, pick application code over an agent. (Inference)

AGENT LOOP WITH MODEL-PLANNED STEPS

An agent plans its own steps and calls tools until the task is complete or a limit stops the run.

Rule Choose the least autonomous pattern that covers every step the task needs.

Trap An agent loop for a task with fixed steps. The agent makes more model calls and varies from run to run with no gain. (Inference)

Exam tip If the steps are known in advance, pick code with a model call at the judgment step, not an agent. (Inference)

▼ RECALL CUES (3)

1. **a fixed task with steps known in advance**
   Answer code with a model call at the judgment step
   Rule Keep the steps in code. Call the model only where judgment is needed.
   - Agent loop for every task
   - Model plan for a fixed sequence
2. **steps unknown until the run starts**
   Answer agent loop with a spend cap
   Rule Use an agent loop when the plan depends on results. Add a budget.
   - Fixed code path that cannot adapt
   - Agent with no budget and no tool list
3. **one judgment step with a side effect**
   Answer tool-use loop in application code
   Rule Define one tool set. Run each tool in your code.
   - Full agent for one judgment step
   - Side effect with no tool result check

▼ NUMBERS AND VERSIONS (1)

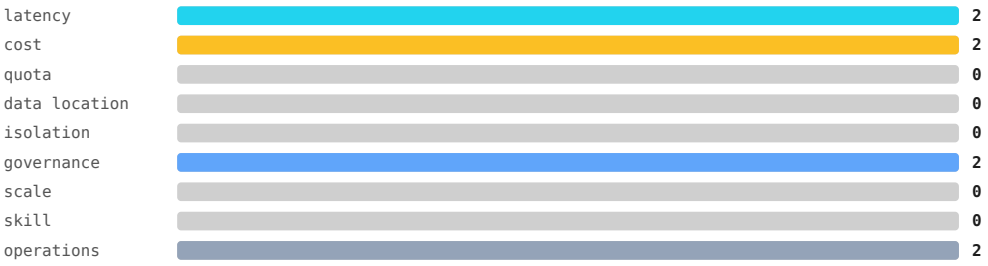
- On Claude Platform on AWS, a Managed Agents session runs without user events for up to 6 hours before it needs reauthentication. source

D1.4 CHAPTER D1, SOLUTION DESIGN AND ARCHITECTURE

Multi-agent orchestration

Independent scoped work only. One lead owns the merge.

CONSTRAINT FAMILIES



SUBAGENT DELEGATION

Each subagent keeps its own conversation history, and the lead agent receives only the subagent's reported results.

Rule Delegate only independent, well-scoped work, and keep one lead agent that owns the merge.

Trap Three agents with no lead and no merge. Each agent returns output, and no one owns one coherent result. (Inference)

Exam tip If the stem needs one coherent result, pick the option with a named lead and a merge step. (Inference)

DYNAMIC WORKFLOW RUNS

Claude writes a program that calls many agents in phases and combines their results without the lead agent in each step.

Rule Use a workflow run for many-piece parallel work, and cap its spend with a session budget.

Trap A chat with each agent in turn. The sequence removes the parallel work that the run could split. (Inference)

Exam tip If the stem asks for a follow-up to each worker, pick subagents, not a workflow run. (Inference)

▼ RECALL CUES (3)

1. **independent specialist work with one deadline**
   Answer lead agent with scoped subagents and a merge
   Rule Split on independent seams. Keep one lead that owns the merge.
   - Three agents with no merge
   - One agent in sequence for independent tracks
2. **many parallel pieces and a progress need**
   Answer workflow run with phases and a budget
   Rule Use a workflow run. Cap spend with a session budget.
   - Chat with each agent in turn
   - Run with no budget
3. **entangled steps that share one context that changes over time**
   Answer single agent
   Rule Keep one agent when each step needs the previous result.
   - Subagents for each step
   - Parallel workers that each need the full history

▼ NUMBERS AND VERSIONS (2)

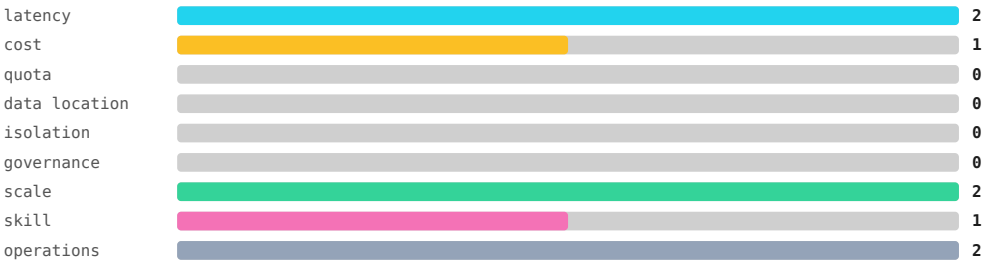
- Delegation is one level deep, and a session can have at most 25 child threads at a time. source
- Multiagent orchestration has beta status with the beta header managed-agents-2026-04-01. source

D1.5 CHAPTER D1, SOLUTION DESIGN AND ARCHITECTURE

Task decomposition

Split at independent seams. Check the merge in code.

CONSTRAINT FAMILIES



FAN-OUT WITH A MERGE STEP

Independent pieces run in parallel, and each piece gets a bounded input and a fixed output format. (Inference)

Rule Split at independent seams, and check the merged result in code.

Trap One large call over the whole input. Long inputs weaken focus on the middle parts, and one error fails the whole run. (Inference)

Exam tip If the stem asks for a check on every number, pick the option with a code check after the merge. (Inference)

CHAIN AND CONTEXT CONTROL

Ordered steps pass each output to the next step, so a check between steps keeps an early error out of later steps. (Inference)

Rule Chain only the steps that need the previous output, and check each output before the next step.

Trap A chain with no check between steps. The first wrong output becomes the input for every later step. (Inference)

Exam tip If the stem names an ordered dependency, pick the chain with a check between steps. (Inference)

▼ RECALL CUES (3)

1. **a long input with independent sections and a traceability need**
   Answer fan-out with citations and a merge check
   Rule Split at section seams. Check the merged totals in code.
   - One large call
   - Chain of unrelated sections
2. **steps that depend on earlier output**
   Answer chain with a check between steps
   Rule Chain only dependent steps. Check each output before the next call.
   - Parallel calls for dependent steps
   - Chain with no check
3. **bulk work with no answer needed at once**
   Answer batch processing
   Rule Use the Message Batches API for bulk work that can wait.
   - Synchronous calls for a large backlog
   - Batch for an interactive user request

▼ NUMBERS AND VERSIONS (5)

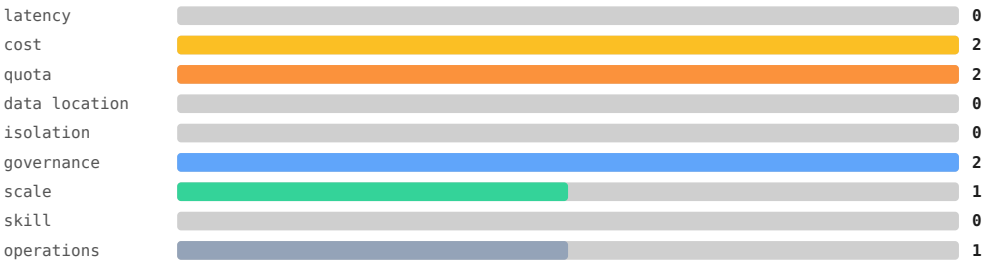
- A batch holds up to 100,000 Message requests or 256 MB, whichever is reached first. source
- Most batches finish within 1 hour. source
- Batch results are available for 29 days after creation. source
- Batch processing gives a 50 percent discount on input and output tokens. source
- A threshold compaction trigger must be at least 50,000 input tokens, and the default trigger is 150,000. source

D1.6 CHAPTER D1, SOLUTION DESIGN AND ARCHITECTURE

Business-value alignment

Unit cost from published rates, times expected volume.

CONSTRAINT FAMILIES



UNIT COST FROM PUBLISHED PRICES

The cost of one request follows from its token counts and the published price per million tokens.

Rule Price the unit cost from the published rates, then multiply it by the expected volume.

Trap A savings claim with one term, such as labor saved, and no build or run cost. The total looks positive and leaves out cost. (Inference)

Exam tip If the stem gives no price, write Not in source and state the price as an assumption. (Inference)

VALUE GUARDS FOR QUALITY AND SPEND

A value claim holds only while its quality threshold and its spend cap still hold. (Inference)

Rule Keep each value claim only while its quality threshold and its spend cap hold.

Trap A latency cut that breaks the quality threshold. The number looks like value, but the error cost moves in the other direction. (Inference)

Exam tip If a stem reports a speed gain, check the quality threshold before you count the gain as value. (Inference)

▼ RECALL CUES (3)

1. **a savings claim with one term**
   Answer all terms priced with numbers
   Rule Count the baseline, the new cost, the errors avoided, and the build and run cost.
   - Savings from labor alone
   - Adjectives such as faster or better with no figure
2. **a model change that lowers cost**
   Answer cost change with a quality threshold
   Rule Keep the cost change only while the eval pass rate holds.
   - Cheapest model with no eval
   - Cost cut with no quality check
3. **a spend overrun risk on long agent runs**
   Answer session budget cap
   Rule Set a session budget before the run starts.
   - Cap set after the bill arrives
   - Lower model tier as the only spend control

▼ NUMBERS AND VERSIONS (4)

- Claude Haiku 5.5 base input price is 0.10 dollars per million tokens for prompts up to 100,000 tokens. source
- Claude Haiku 5.5 output price is 0.50 dollars per million tokens for prompts up to 100,000 tokens. source
- Claude Haiku 5.5 cache hits cost 0.01 dollars per million tokens. source
- Toy: 2 million input tokens cost 0.20 dollars at the Haiku 5.5 base input price of 0.10 dollars per million. (Inference) source